WEB DEVELOPMENT PROJECT REPORT

TechZone Electronics – Multi-Page E-Commerce Website

---

1. Introduction

In the modern digital era, an online web presence is critical for businesses in the retail and electronics industry. This project presents the design and development of 'TechZone Electronics', a clean, multi-page, responsive web platform designed to showcase technological products, provide categorical browsing, and allow prospective buyers to compare technical specifications and pricing side-by-side. The website incorporates modern layout patterns including CSS Flexbox, semantic HTML5, and responsive design principles to deliver a seamless shopping experience across desktop and mobile devices.

2. Aim

To design and implement a multi-page, responsive electronics catalog website titled 'TechZone Electronics' using semantic HTML5 and CSS3, featuring a structured product catalog, promotional hero banner, and an interactive specifications comparison interface.

3. Objectives

- Develop a coherent multi-page website architecture consisting of Home (`index.html`), Products (`products.html`), and Product Comparison (`compare.html`).
- Implement consistent brand styling, typographic hierarchy, and intuitive navigation across all pages using an external stylesheet (`style.css`).
- Utilize semantic HTML5 structural tags such as `<header>`, `<nav>`, `<section>`, `<table>`, and `<footer>` for improved document accessibility and SEO.
- Showcase electronic items across categorized sections (Laptops, Smartphones, Accessories, Smart Devices) using card layouts, badges, and image cards.
- Build comparative specification tables with alternating tabular data and clear header formatting to assist users in technical purchase decisions.
- Implement CSS Media Queries (`@media (max-width: 600px)`) to guarantee responsive layouts on mobile devices.

4. HTML Concepts Used

The project leverages several fundamental and modern HTML5 concepts to build well-structured and accessible web pages:

- **Semantic Layout Elements:** Use of `<header>`, `<nav>`, `<section>`, and `<footer>` tags to provide logical landmark segmentation of the document.
- **Hypertext Navigation & Anchor Linking:** Use of `<a href="...">` tags for inter-page navigation between pages as well as intra-page bookmark jump links (e.g., `#laptops`, `#phones`, `#accessories`, `#smart`).
- **Media Integration:** Use of `<img>` tags with standard image sources, `alt` attributes for screen readers and accessibility, and fluid responsive widths.
- **Tabular Data Representation:** Implementation of `<table>`, `<tr>`, `<th>`, and `<td>` tags to present side-by-side comparative product specifications (CPU, RAM, Storage, OS, Display, and Price).
- **Visual Badging & Categorization:** Application of inline grouping tags such as `<span>` and `<div class="card">` for pricing tags, offer discounts, and modular card wrappers.

5. CSS Concepts Used

Styling is handled via an external stylesheet (`style.css`) that establishes a cohesive color palette, typography, and adaptive layouts:

- **Box Model & Global Reset:** Applied `* { margin: 0; padding: 0; box-sizing: border-box; }` to establish predictable element dimensions and eliminate default browser spacing differences.
- **Color Palette & Visual Hierarchy:** Deep navy blue (`#172554`) for headers/footers, royal blue (`#2563eb`) for accent bars and buttons, soft ice-blue (`#dbeafe`) for banner background, light gray (`#f4f6f8`) body canvas, emerald green (`#15803d`) for prices, and vivid crimson (`#dc2626`) for discount badges.
- **CSS Flexbox Layout:** Applied `display: flex; flex-wrap: wrap; justify-content: center; gap: 25px;` on `.container` elements to automatically arrange product cards into flexible responsive grids without float hacks.
- **Component Styling & Box Shadows:** Styled product cards (`.card`) with subtle rounded borders (`border-radius: 10px`), clean padding, and soft box-shadows (`0 2px 8px #cccccc`) to give depth.
- **Table Styling:** Full-width table formatting with `border-collapse: collapse`, custom cell padding (`15px`), crisp light gray cell borders (`#cccccc`), and colored header rows.
- **Interactive Pseudo-Classes:** Hover effects such as `nav a:hover { color: yellow; }` and `.filters a:hover { background-color: #172554; }` providing visual feedback.
- **Media Queries for Responsiveness:** `@media (max-width: 600px)` breakpoint adapting card widths to `90%`, adjusting table font sizes and padding, and modifying navigation layout for handheld touch devices.

6. Procedure

Step 1: Requirement Analysis & Planning: Defined the site structure, identifying the need for three interconnected pages (Home, Products, Compare) and establishing the product catalog data requirements.

Step 2: Wireframing & Theme Definition: Selected the TechZone blue/navy color scheme, typography (Arial/sans-serif), and responsive card-based layout structure.

Step 3: Creating the Shared Stylesheet (style.css): Wrote centralized CSS rules including universal reset, header/nav formatting, banner sections, card component styling, tables, and mobile media queries.

Step 4: Developing the Home Page (index.html): Constructed the brand header, top navigation menu, promotional call-to-action banner, featured product grid, value proposition section, and shared footer.

Step 5: Developing the Products Catalog (products.html): Implemented anchor-based category filter buttons and arranged products under four categorized sections (Laptops, Smartphones, Accessories, Smart Devices).

Step 6: Developing the Comparison Page (compare.html): Implemented side-by-side data tables for laptops and smartphones comparing hardware specifications, OS, prices, and discounts.

Step 7: Testing & Responsive Verification: Verified link integrity, checked multi-page navigation flows, and tested display resizing across desktop and mobile screen viewports.

7. Source Code

7.1 index.html (Home Page)

```
<!DOCTYPE html>
<html>
<head>
    <title>TechZone Electronics</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <header>
        <h1>TechZone Electronics</h1>
        <p>Your One-Stop Electronics Store</p>
    </header>

    <nav>
        <a href="index.html">Home</a>
        <a href="products.html">Products</a>
        <a href="compare.html">Compare Products</a>
    </nav>

    <section class="banner">
        <h2>Upgrade Your Technology</h2>
        <p>Discover laptops, smartphones and smart devices.</p>
```

```
        <a class="btn" href="products.html">Shop Now</a>
    </section>

    <h2 class="heading">Featured Products</h2>

    <div class="container">

        <div class="card">
            <img src="https://images.unsplash.com/photo-1496181133206-80ce9b88a853?w=500"
alt="Laptop">
            <h3>HP Laptop</h3>
            <p>8GB RAM, 512GB SSD</p>
            <p class="price">₹45,999</p>
            <span class="discount">10% OFF</span>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1511707171634-5f897ff02aa9?w=500"
alt="Smartphone">
            <h3>Smartphone</h3>
            <p>128GB Storage, 8GB RAM</p>
            <p class="price">₹19,999</p>
            <span class="discount">15% OFF</span>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=500"
alt="Headphones">
            <h3>Wireless Headphones</h3>
            <p>Bluetooth connectivity</p>
            <p class="price">₹1,999</p>
            <span class="discount">20% OFF</span>
        </div>

    </div>

    <section class="info">
        <h2>Why Shop With Us?</h2>
        <p>✓ Quality Products</p>
        <p>✓ Affordable Prices</p>
        <p>✓ Latest Technology</p>
        <p>✓ Wide Range of Electronics</p>
    </section>

    <footer>
        <p>© 2026 TechZone Electronics</p>
    </footer>

</body>
</html>
```

7.2 products.html (Products Catalog)

```
<!DOCTYPE html>
<html>
<head>
    <title>Our Products - TechZone</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <header>
        <h1>Our Products</h1>
        <p>Explore Our Electronics Collection</p>
    </header>

    <nav>
        <a href="index.html">Home</a>
        <a href="products.html">Products</a>
        <a href="compare.html">Compare Products</a>
    </nav>

    <h2 class="heading">Product Categories</h2>

    <div class="filters">
        <a href="#laptops">Laptops</a>
        <a href="#phones">Smartphones</a>
        <a href="#accessories">Accessories</a>
        <a href="#smart">Smart Devices</a>
    </div>

    <h2 class="heading" id="laptops">Laptops</h2>
    <div class="container">

        <div class="card">
            <img src="https://images.unsplash.com/photo-1496181133206-80ce9b88a853?w=500"
alt="Laptop">
            <h3>HP Laptop</h3>
            <p>Processor: Intel Core i5</p>
            <p>RAM: 8GB</p>
            <p>Storage: 512GB SSD</p>
            <p class="price">₹45,999</p>
            <span class="discount">10% OFF</span>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1517336714731-489689fd1ca8?w=500"
alt="Silver laptop">
            <h3>Apple MacBook</h3>
            <p>Processor: Apple M-series</p>
            <p>RAM: 8GB</p>
            <p>Storage: 256GB SSD</p>
            <p class="price">₹79,999</p>
            <span class="discount">5% OFF</span>
```

```
        </div>

    </div>

    <h2 class="heading" id="phones">Smartphones</h2>
    <div class="container">

        <div class="card">
            <img src="https://images.unsplash.com/photo-1511707171634-5f897ff02aa9?w=500"
alt="Smartphone">
            <h3>Android Smartphone</h3>
            <p>RAM: 8GB</p>
            <p>Storage: 128GB</p>
            <p>Display: AMOLED</p>
            <p class="price">₹19,999</p>
            <span class="discount">15% OFF</span>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1592750475338-74b7b21085ab?w=500"
alt="Mobile phone">
            <h3>Premium Smartphone</h3>
            <p>RAM: 8GB</p>
            <p>Storage: 256GB</p>
            <p>Display: OLED</p>
            <p class="price">₹29,999</p>
            <span class="discount">10% OFF</span>
        </div>

    </div>

    <h2 class="heading" id="accessories">Accessories</h2>
    <div class="container">

        <div class="card">
            <img src="https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=500"
alt="Headphones">
            <h3>Wireless Headphones</h3>
            <p>Bluetooth connectivity</p>
            <p>Built-in microphone</p>
            <p class="price">₹1,999</p>
            <span class="discount">20% OFF</span>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1527814050087-3793815479db?w=500"
alt="Computer mouse">
            <h3>Wireless Mouse</h3>
            <p>Wireless connectivity</p>
            <p>Ergonomic design</p>
            <p class="price">₹799</p>
            <span class="discount">10% OFF</span>
        </div>
```

```
    </div>

    <h2 class="heading" id="smart">Smart Devices</h2>
    <div class="container">

        <div class="card">
            <img src="https://images.unsplash.com/photo-1523275335684-37898b6baf30?w=500"
alt="Smartwatch">
            <h3>Smartwatch</h3>
            <p>Fitness tracking</p>
            <p>Heart-rate monitoring</p>
            <p class="price">₹2,499</p>
            <span class="discount">15% OFF</span>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1608043152269-423dbba4e7e1?w=500"
alt="Smart speaker">
            <h3>Smart Speaker</h3>
            <p>Wireless audio</p>
            <p>Bluetooth connectivity</p>
            <p class="price">₹3,499</p>
            <span class="discount">10% OFF</span>
        </div>

    </div>

    <footer>
        <p>© 2026 TechZone Electronics</p>
    </footer>

</body>
</html>
```

7.3 compare.html (Product Comparison)

```
<!DOCTYPE html>
<html>
<head>
    <title>Compare Products - TechZone</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <header>
        <h1>Compare Products</h1>
        <p>Choose the Right Product for You</p>
    </header>

    <nav>
```

```
        <a href="index.html">Home</a>
        <a href="products.html">Products</a>
        <a href="compare.html">Compare Products</a>
    </nav>

    <h2 class="heading">Laptop Comparison</h2>

    <table>
        <tr>
            <th>Specifications</th>
            <th>HP Laptop</th>
            <th>Apple MacBook</th>
        </tr>

        <tr>
            <td>Image</td>
            <td>
                <img class="table-img" src="https://images.unsplash.com/photo-1496181133206-
80ce9b88a853?w=300" alt="HP laptop">
            </td>
            <td>
                <img class="table-img" src="https://images.unsplash.com/photo-1517336714731-
489689fd1ca8?w=300" alt="MacBook">
            </td>
        </tr>

        <tr>
            <td>Processor</td>
            <td>Intel Core i5</td>
            <td>Apple M-series</td>
        </tr>

        <tr>
            <td>RAM</td>
            <td>8GB</td>
            <td>8GB</td>
        </tr>

        <tr>
            <td>Storage</td>
            <td>512GB SSD</td>
            <td>256GB SSD</td>
        </tr>

        <tr>
            <td>Operating System</td>
            <td>Windows</td>
            <td>macOS</td>
        </tr>

        <tr>
            <td>Price</td>
            <td>₹45,999</td>
```

```
            <td>₹79,999</td>
        </tr>

        <tr>
            <td>Discount</td>
            <td>10% OFF</td>
            <td>5% OFF</td>
        </tr>
    </table>

    <h2 class="heading">Smartphone Comparison</h2>

    <table>
        <tr>
            <th>Specifications</th>
            <th>Android Phone</th>
            <th>Premium Phone</th>
        </tr>

        <tr>
            <td>RAM</td>
            <td>8GB</td>
            <td>8GB</td>
        </tr>

        <tr>
            <td>Storage</td>
            <td>128GB</td>
            <td>256GB</td>
        </tr>

        <tr>
            <td>Display</td>
            <td>AMOLED</td>
            <td>OLED</td>
        </tr>

        <tr>
            <td>Price</td>
            <td>₹19,999</td>
            <td>₹29,999</td>
        </tr>
    </table>

    <footer>
        <p>© 2026 TechZone Electronics</p>
    </footer>

</body>
</html>
```

7.4 style.css (Shared Stylesheet)

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f4f6f8;
    color: #333;
}

header {
    background-color: #172554;
    color: white;
    text-align: center;
    padding: 25px;
}

header h1 {
    margin-bottom: 10px;
}

nav {
    background-color: #2563eb;
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
    font-weight: bold;
}

nav a:hover {
    color: yellow;
}

.banner {
    background-color: #dbeafe;
    text-align: center;
    padding: 70px 20px;
}

.banner h2 {
    font-size: 32px;
    margin-bottom: 15px;
}
```

```
.banner p {
    margin-bottom: 25px;
}

.btn {
    background-color: #2563eb;
    color: white;
    padding: 12px 25px;
    text-decoration: none;
    border-radius: 5px;
}

.heading {
    text-align: center;
    margin: 30px 0 20px;
}

.container {
    display: flex;
    flex-wrap: wrap;
    justify-content: center;
    gap: 25px;
    padding: 20px;
}

.card {
    background-color: white;
    width: 280px;
    padding: 15px;
    text-align: center;
    border-radius: 10px;
    box-shadow: 0 2px 8px #cccccc;
}

.card img {
    width: 100%;
    height: 180px;
    object-fit: cover;
    border-radius: 8px;
}

.card h3 {
    margin: 15px 0;
}

.card p {
    margin: 10px 0;
}

.price {
    color: #15803d;
    font-weight: bold;
    font-size: 20px;
```

```
}

.discount {
    display: inline-block;
    background-color: #dc2626;
    color: white;
    padding: 6px 10px;
    border-radius: 5px;
    margin-top: 5px;
}

.filters {
    text-align: center;
    margin: 20px;
}

.filters a {
    display: inline-block;
    background-color: #2563eb;
    color: white;
    text-decoration: none;
    padding: 10px 15px;
    margin: 5px;
    border-radius: 5px;
}

.filters a:hover {
    background-color: #172554;
}

.info {
    background-color: white;
    text-align: center;
    padding: 30px;
    margin: 20px;
}

.info p {
    margin: 12px;
}

table {
    width: 90%;
    margin: 20px auto 40px;
    border-collapse: collapse;
    background-color: white;
}

th, td {
    border: 1px solid #cccccc;
    padding: 15px;
    text-align: center;
}
```

```
th {
    background-color: #2563eb;
    color: white;
}

.table-img {
    width: 120px;
    height: 90px;
    object-fit: cover;
}

footer {
    background-color: #172554;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 30px;
}

@media (max-width: 600px) {
    nav a {
        display: inline-block;
        margin: 8px;
    }

    .banner h2 {
        font-size: 25px;
    }

    .card {
        width: 90%;
    }

    table {
        width: 100%;
        font-size: 12px;
    }

    th, td {
        padding: 8px;
    }

    .table-img {
        width: 65px;
        height: 60px;
    }
}
```

8. Output Screenshots & UI Description

The multi-page website outputs are divided into three core user interfaces corresponding to each HTML page:

8.1 Home Page (index.html) Interface

- Header & Branding: Deep navy banner displaying 'TechZone Electronics' and tagline 'Your One-Stop Electronics Store'.
- Navigation Bar: Blue horizontal bar containing active navigation links with hover color changes.
- Hero Banner: Soft light blue callout box highlighting 'Upgrade Your Technology' with a rounded 'Shop Now' call-to-action button.
- Featured Products: Flexbox container featuring cards for HP Laptop (₹45,999, 10% OFF), Smartphone (₹19,999, 15% OFF), and Wireless Headphones (₹1,999, 20% OFF).
- Why Shop With Us: Centered card highlighting key value propositions (Quality Products, Affordable Prices, Latest Technology, Wide Range).
- Footer: Centered copyright notice on dark navy background.

8.2 Products Catalog Page (products.html) Interface

- Category Pill Navigation: Interactive button row linking to #laptops, #phones, #accessories, and #smart.
- Laptops Section: Side-by-side cards comparing the HP Laptop (Intel Core i5, 512GB SSD) and Apple MacBook (M-series, 256GB SSD).
- Smartphones Section: Side-by-side cards for Android Smartphone (128GB, AMOLED) and Premium Smartphone (256GB, OLED).
- Accessories Section: Cards displaying Wireless Headphones (₹1,999) and Wireless Mouse (₹799).
- Smart Devices Section: Cards displaying Smartwatch (₹2,499) and Smart Speaker (₹3,499).

8.3 Comparison Page (compare.html) Interface

- Laptop Comparison Table: Clean matrix comparing Processor, RAM, Storage, OS, Price, and Discounts with product preview thumbnails.
- Smartphone Comparison Table: Structured comparison across RAM, Storage, Display, and Price.
- Table Styling: Royal blue column headers, clear grid lines, and centered specifications.

Interface Page	Visual Screen Capture / Representation
Home Page (index.html)	*[Attach Screenshot: Hero banner, navigation, featured product cards, and value points]*
Products Catalog (products.html)	*[Attach Screenshot: Category anchors, grouped grid of product cards with pricing badges]*
Product Comparison (compare.html)	*[Attach Screenshot: Tabular layout showing laptop and*

smartphone comparison tables]

9. Conclusion

The 'TechZone Electronics' web development project successfully fulfilled all foundational requirements of a modern e-commerce catalog and comparison website. By structuring clean semantic HTML5 markup across multiple files and linking them via a unified CSS stylesheet, the project maintains separation of concerns, high maintainability, and visual consistency. The implementation of CSS Flexbox enables smooth layout adaptability across various resolutions, while the comparison tables provide an intuitive mechanism for consumers to evaluate device specifications before purchasing. This project reinforces practical understanding of semantic web development, UI component styling, responsive web design principles, and navigation architecture.